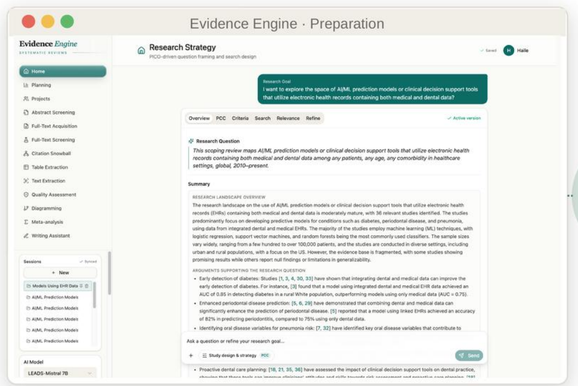
Evidence Engine: An Open-Source, Human-in-the-Loop Platform for LLM-Assisted Evidence Synthesis and Reviews

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Workflow Stages



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STAGE 1 · PREPARATION

Research strategy

A natural language goal becomes a structured review framework, eligibility criteria, and a Boolean/MeSH query, with a cited summary of relevant studies.

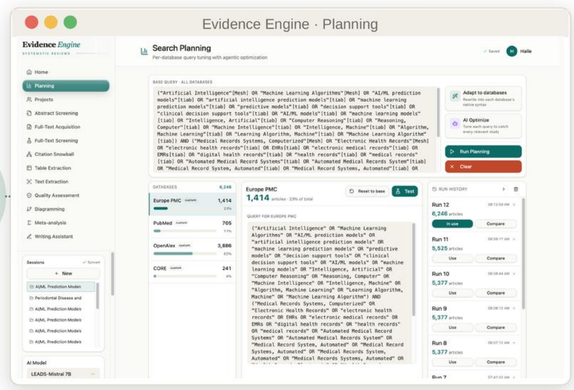
Home · Projects

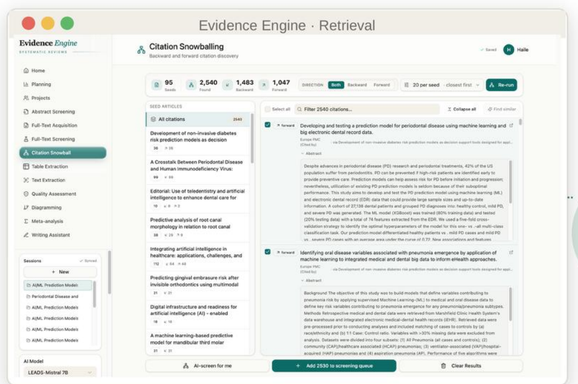
STAGE 2 · PLANNING

Search planning

One base query is rewritten into each search engine's syntax with projected yield before retrieval.

Planning





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STAGE 3 · RETRIEVAL

Article retrieval and snowballing

Federated search with de-duplication and full-text retrieval; backward and forward citation snowballing expands the set from seed papers.

Full-Text Acquisition · Citation Snowball

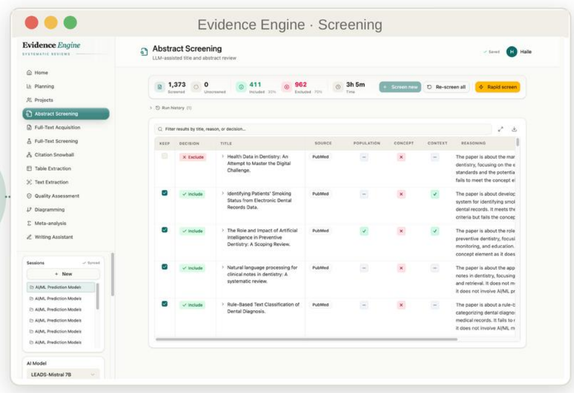
STAGE 4 · SCREENING

Abstract and full-text screening

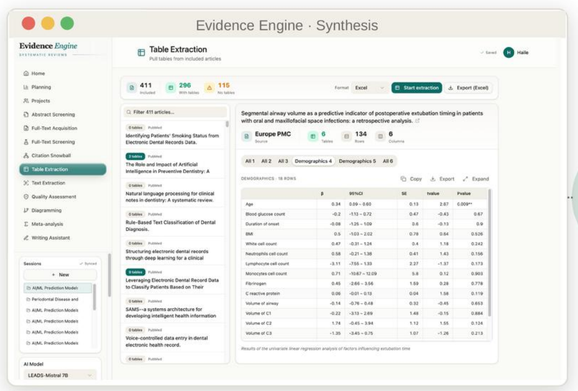
A four-way verdict per framework element with its reasoning, for abstracts and full texts where every decision stays overridable.

$$s(v) = \begin{cases} +1 & \text{YES} \\ +\tfrac{1}{2} & \text{PARTIAL} \\ 0 & \text{UNCERTAIN} \\ -1 & \text{NO} \end{cases} \qquad a_i = \frac{1}{|E|}\sum_{e \in E} s(v_{i,e}) \in [-1, 1] \qquad \hat{y}_i = \mathbb{1}\left[a_i \geq \tau\right]$$

Abstract Screening · Full-Text Screening



4



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STAGE 5 · SYNTHESIS

Appraisal and extraction

Tables are parsed deterministically, so their values cannot be generated; extraction forms leverage LLMs to tie every answer to a quoted source. Both export straight to Excel.

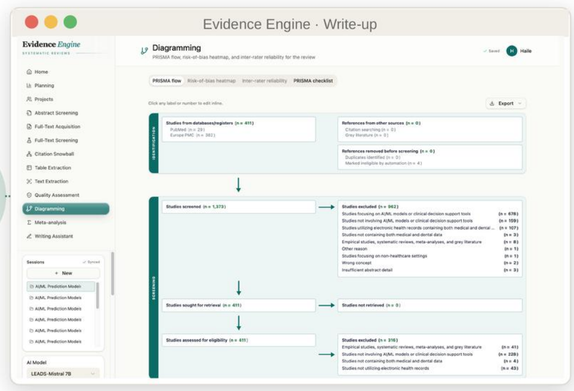
Table Extraction · Text Extraction · Quality Assessment · Meta-analysis

STAGE 6 · WRITE-UP

PRISMA reporting

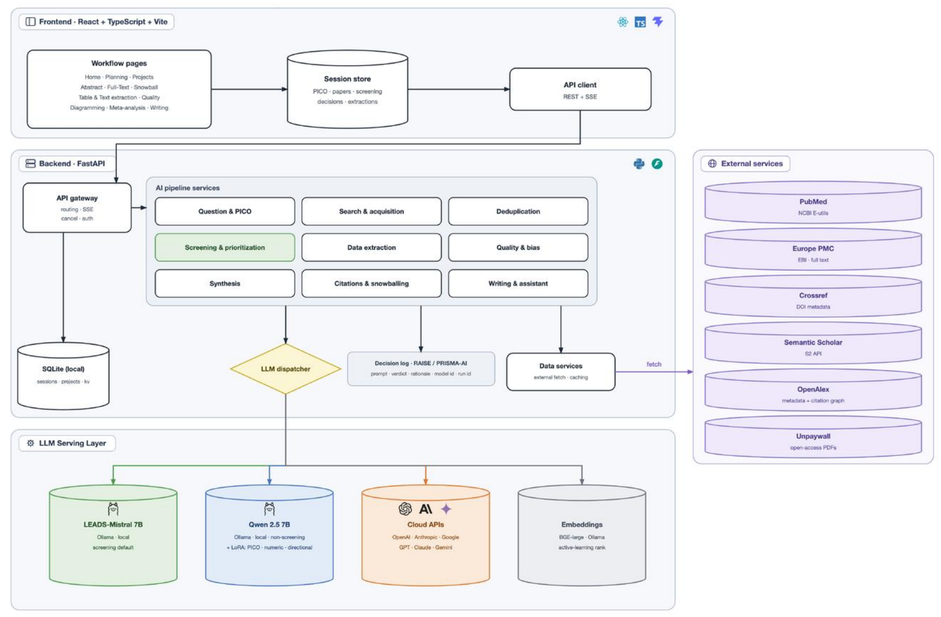
PRISMA-2020 flow diagram with every exclusion reason, exported for the manuscript; a writing assistant brings the question, search, screening decisions and extracted data together into a cited draft.

Diagramming · Writing Assistant

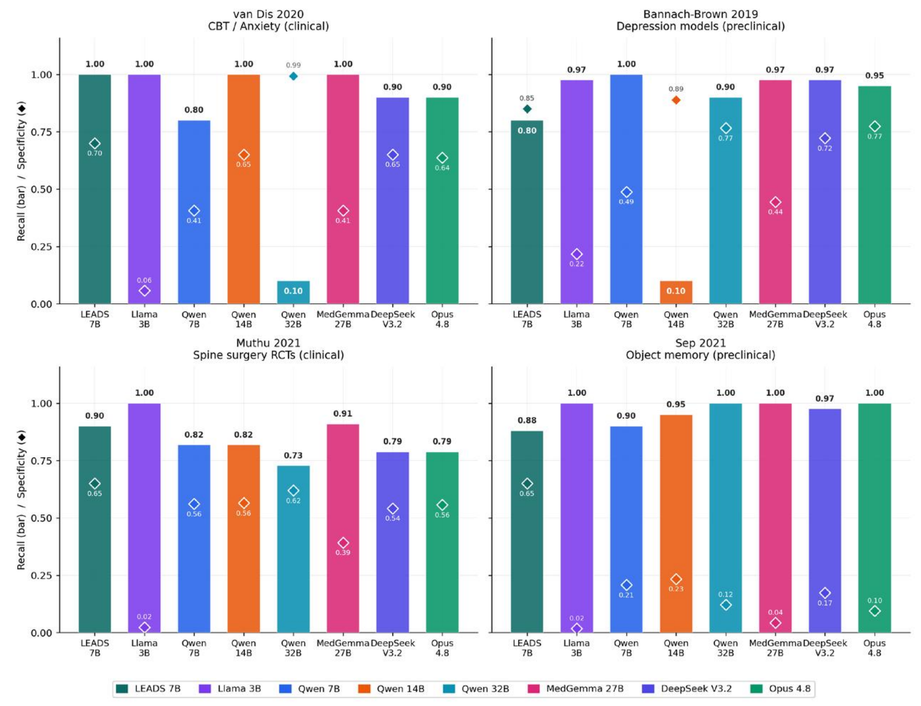


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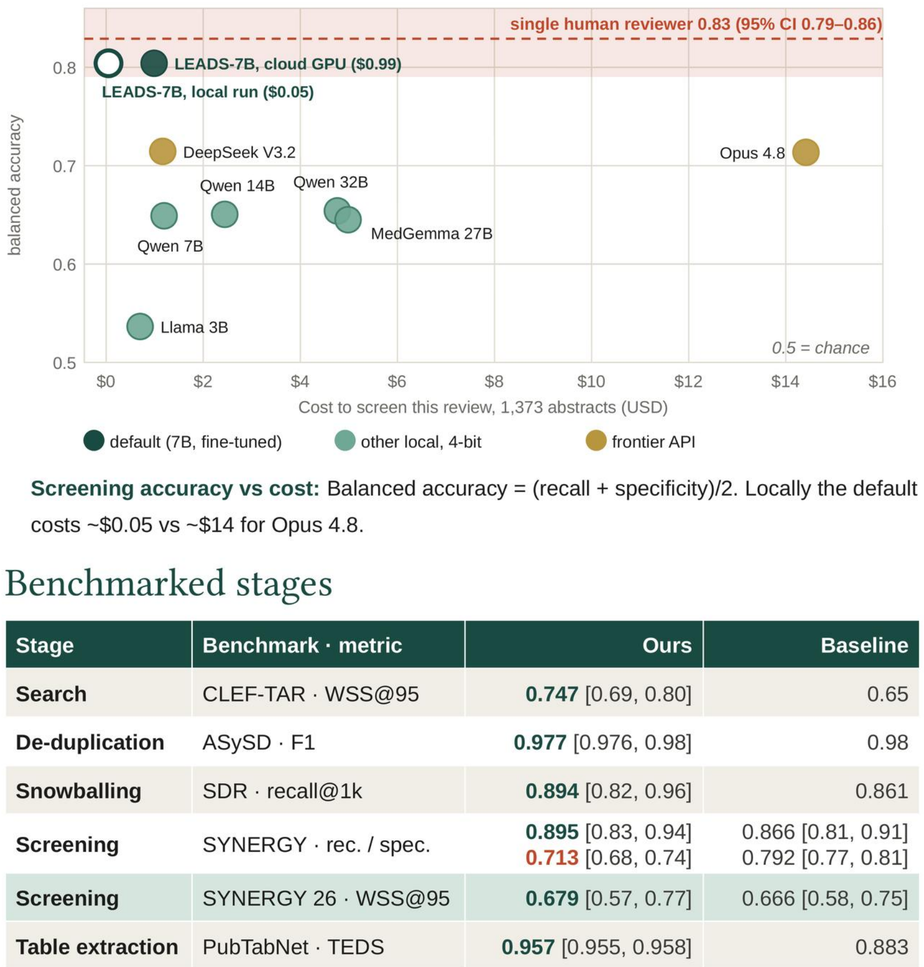
Architecture



Results



By model and dataset: Recall (bars), specificity (diamonds). Across 26 SYNERGY reviews with no labels: recall 0.986, specificity 0.61, AUROC 0.849. Work saved (WSS@95) vs ASReview: 0.68 vs 0.67 with active learning; 0.65 vs 0.68 label-free.



Screening accuracy vs cost: Balanced accuracy = (recall + specificity)/2. Locally the default costs ~$0.05 vs ~$14 for Opus 4.8.

Benchmarked stages

Stage	Benchmark · metric	Ours	Baseline
Search	CLEF-TAR · WSS@95	**0.747** [0.69, 0.80]	0.65
De-duplication	ASySD · F1	**0.977** [0.976, 0.98]	0.98
Snowballing	SDR · recall@1k	**0.894** [0.82, 0.96]	0.861
Screening	SYNERGY · rec. / spec.	**0.895** [0.83, 0.94] **0.713** [0.68, 0.74]	0.866 [0.81, 0.91] 0.792 [0.77, 0.81]
Screening	SYNERGY 26 · WSS@95	**0.679** [0.57, 0.77]	0.666 [0.58, 0.75]
Table extraction	PubTabNet · TEDS	**0.957** [0.955, 0.958]	0.883
Numeric data	Yun 2024 · exact match	**0.604** [0.52, 0.68]	0.655
Risk of bias	ROBoto2 · micro-F1	**0.534** [0.51, 0.56]	0.61

Conclusions

- A local 7B model matches frontier recall (0.895 vs 0.909) for a fraction of cost.
- Orchestration across the pipeline provides comparable performance across published baselines on most stages.
- Domain tuning often beats scale.
- Misses are recoverable, since most missed includes was flagged UNCERTAIN or PARTIAL, never NO.
- Next, we will compare prospectively against human teams, support living reviews, and extend beyond biomedicine.

Contact Haile Teshome · haile.teshome@ucsf.edu **Code & data** github.com/haile-teshome/Evidence-Engine (MIT) **Paper** LM4Sci workshop @ COLM 2026